VoxNeuro: Leakage-Safe Parkinson's Screening Support via Multi-View Grassmannian Speech Analysis

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Abstract: Parkinson's disease (PD) alters phonation and articulation, making repeated speech a candidate digital biomarker. Public speech datasets contain several recordings per participant, and record-wise cross-validation can place one person in both training and test sets, inflating reported scores. VoxNeuro treats the subject as the observation. Each subject's recordings are rank-normalized with training-fold functions and, when features outnumber training recordings, projected onto supervised partial least squares subspaces; they then define a nominal rank-three Grassmann subspace and a Euclidean mean–dispersion vector, whose Gaussian kernels are fused into a positive-semidefinite kernel for support vector classification. Under five-fold subject-disjoint evaluation on two public cohorts, VoxNeuro reached balanced accuracies of 0.875 (UCI-489) and 0.816 (PD-252), higher than all five standard comparators (including a class-weighted random forest and PLS discriminant analysis) under identical folds and normalization, with PD-252 sensitivity 0.899 and specificity 0.734. Across 20 further partitions, its PD-252 estimates were higher than every standard comparator in at least 19 and than a z-standardized full-space model in all 20; ablations associate the difference with rank normalization and the supervised-subspace branch, whereas the UCI-489 margin over a full-space Euclidean-kernel ablation was not partition-consistent. Design choices were fixed on one partition; no external validation was performed.

Keywords: Parkinson's disease; speech biomarker; Grassmann manifold; kernel fusion; imbalanced learning; subject-disjoint cross-validation; support vector machine; rank normalization; partial least squares; digital health

1. Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder involving nigrostriatal dopaminergic loss, basal ganglia circuit dysfunction, and heterogeneous motor and non-motor manifestations [1–3]. Its diagnosis remains clinical [4,5], and its global burden has grown faster than that of any other neurological disorder [6,7]. Disordered speech is among the most common motor consequences: dysarthria, hypophonia, and related vocal-tract dysfunctions affect the large majority of patients over the disease course [8,9]. Quantitative acoustic analysis detects these changes even in early and untreated disease [10]. Nonlinear dysphonia measures separate disordered from healthy voices [11], and repeated noninvasive speech tests have supported research on low-burden remote monitoring and progression tracking [12–14]. Automated speech analysis has revealed prodromal changes in at-risk cohorts [15]. Machine-learning approaches to voice-based PD detection have accordingly expanded rapidly, as documented by recent systematic reviews [16,17]. A digital biomarker, however, is clinically credible only when its evaluation matches its intended unit of use: the patient.

Public PD speech datasets often contain multiple recordings per participant. When recordings are shuffled independently, subject-specific anatomical and acquisition signatures can occur on both sides of a record-wise split, and the evaluation then estimates recognition of another recording from a previously observed person rather than generalization to an unseen person. This failure mode is a specific instance of data leakage, a recognized driver of over-optimistic results across applied machine learning [18,19]. This effect has been demonstrated directly in clinical prediction: record-wise cross-validation can inflate accuracy substantially relative to subject-wise evaluation of the very same pipeline [20], a finding that prompted an extended methodological discussion [21]. Subject characteristics can also drive apparent performance even when splits look innocuous [22]. Related biases arise whenever model selection and error estimation share data [23,24], and cross-validation on small samples carries wide error bars even under favorable conditions [25–27]. The UCI-489 corpus analyzed here contains three dependent replications from each of 80 subjects [28]; replication-aware analysis has been proposed for exactly this cohort [29].

Methodologically, three research strands converge in this study. First, engineered speech descriptors: dysphonia, cepstral, wavelet, and tunable Q-factor wavelet transform (TQWT) descriptor families have been used to characterize PD-related voice changes [12,30,31]. Second, subspace geometry: collections of related observations can be represented as points on the Grassmann manifold $\mathrm{Gr}(r, d)$, whose quotient structure removes arbitrary basis choices [32–34]; this representation underpins subspace-based learning in computer vision and pattern recognition [35,36], and kernels on such manifolds require care because Gaussian functions of arbitrary manifold distances are not automatically positive semidefinite (PSD) [37]. Third, kernel fusion: reproducing-kernel Hilbert space (RKHS) theory [38–40] and multiple-kernel learning [41–43] combine heterogeneous similarities while preserving convex support vector optimization. Class imbalance, finally, is routinely addressed by synthetic oversampling [44–46], whose behavior in high-dimensional feature spaces warrants caution [47]. This study combines these strands under an evaluation protocol that treats the patient as the statistical unit.

Deep architectures, including convolutional, recurrent, and transformer-based models, are increasingly applied to voice-based PD detection and scale well when large corpora are available [16]. Smartphone-scale collection efforts such as mPower point in that direction [48,49], and technology roadmaps for PD anticipate exactly this kind of remote, high-frequency assessment [50–52]. The cohorts relevant to replicated-phonation screening, however, typically contain tens to a few hundred subjects. In this regime, the present study deliberately combines structured geometric priors with a convex, auditable classical learner instead of high-capacity end-to-end models: each component (representation, kernel, solver, and resampling step) has an explicit mathematical description that can be verified directly against its implementation.

VoxNeuro aligns the mathematical unit of analysis with the clinical unit of interest: each subject is represented by the complete set of repeated measurements. A fixed-rank representation based on singular value decomposition (SVD) encodes the joint orientation of the repetitions, while a Euclidean summary preserves feature-space location and within-subject variability. Both views are built after fold-local rank normalization and, in the high-dimensional regime, inside a supervised low-dimensional subspace; a positive-semidefinite direct-sum kernel combines them, and evaluation is subject-disjoint throughout (Figure 1). For imbalanced full-space training folds, synthetic subjects are generated along manifold geodesics rather than by interpolating individual recording rows. The study had three objectives: (i) to formalize why record-wise and subject-disjoint evaluation estimate different risks; (ii) to assess whether the resulting subject-level classifier improves on standard comparators evaluated under the same leakage-safe protocol; and (iii) to attribute its performance to its components through ablations. A separate research-stage web prototype for guided speech capture is described in the Discussion but was not used in any analysis.

The principal contributions are as follows: (i) a formal subject-disjoint risk definition together with a quantitative account of the optimistic bias induced by record-wise splitting; (ii) a two-view kernel construction in which a basis-invariant Grassmann representation of repeated speech supplies a mathematically valid but empirically modest complementary term to a Euclidean summary, fused through a provably positive-semidefinite kernel; (iii) fold-local Gaussian rank normalization and a dimension-adaptive supervised-subspace stage that is activated only when the feature count exceeds the number of training recordings; and (iv) an implemented conditional branch of geodesic minority oversampling for imbalanced full-space training folds, which is exercised by the full-space comparators but remains dormant in the two reported evaluations of the complete model. One shared configuration is applied to both cohorts, the model is compared with standard comparators (class-weighted logistic regression, a class-weighted linear SVM, SMOTE-based logistic regression, a class-weighted random forest, and PLS discriminant analysis) and with a full-space Euclidean-kernel ablation under identical folds and normalization, and every component is ablated. The configuration was fixed after inspection of one reference partition, and its sensitivity to the subject partition is reported separately.

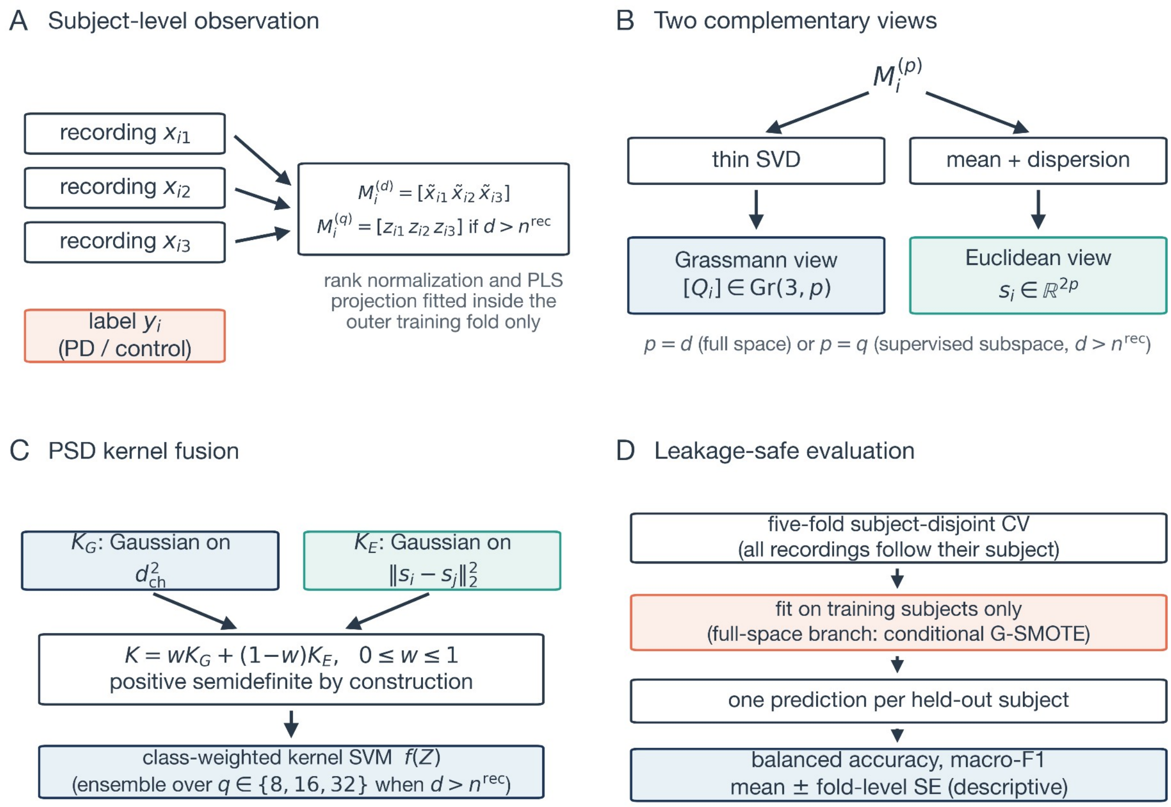

Figure 1. Schematic overview of VoxNeuro. (**A**) Each subject contributes three repeated recordings and one diagnostic label; the recordings are normalized with functions fitted inside the outer training fold only (Gaussian rank normalization) and stacked into a matrix M_i; when the feature count exceeds the number of training recordings, the recordings are first projected onto a supervised subspace (Section 2.7). (**B**) A thin singular value decomposition (SVD) yields the Grassmann view $[Q_i]$, while coordinate-wise mean and dispersion yield the Euclidean view s_i; the views live in $\mathbb{R}^p$ with $p = d$ in the full space and $p = q$ in the supervised subspace. (**C**) Gaussian kernels on the two views are combined into a positive-semidefinite (PSD) fused kernel feeding a class-weighted support vector machine (SVM). (**D**) Evaluation is five-fold subject-disjoint cross-validation (CV); in the full-space branch, conditional Grassmann-geodesic synthetic minority oversampling (G-SMOTE) is activated only inside imbalanced training folds, and fold scores are summarized as mean ± standard error (SE). PD, Parkinson's disease.

2. Materials and Methods

2.1. Datasets

Two public, de-identified cohorts were analyzed (Table 1). The first, denoted UCI-489, comprises 80 subjects (40 PD, 40 control) with three recordings of sustained /a/ per subject, each recording described by 44 replicated acoustic descriptors and one dataset-provided binary variable labeled gender [28,29]. The second, denoted PD-252 [53], comprises 252 subjects (188 PD, 64 control) with three recordings per subject, described by 752 descriptors spanning baseline acoustic, vocal/time-frequency, Mel-frequency cepstral coefficient (MFCC) [54], wavelet, TQWT [55], and intrinsic mode function/empirical mode decomposition (IMF/EMD) [56] families [30,31]. It also includes one dataset-provided binary variable labeled gender, which constitutes the demographic family of the permutation analysis in Appendix A.2; neither source dataset distinguishes sex from gender for this variable. The coded variable was included as an input feature and was never used to construct folds. The cohorts were selected as complementary evaluation regimes for the same protocol: on balanced, lower-dimensional UCI-489 the full-space branch of the model is active and no synthetic subjects are generated, whereas on the larger, high-dimensional, imbalanced PD-252 cohort the dimensionality gate of Section 2.7 activates the supervised-subspace branch, and the full-space comparators exercise the imbalance-handling branch. This study was a secondary computational analysis of public data and involved no participant recruitment.

Table 1. Dataset regimes and evaluation design.

Attribute	UCI-489	PD-252
Subjects	80	252
PD / control	40 / 40	188 / 64
Recordings per subject	3	3
Modeled variables	45	753
Recording task	Sustained /a/, three repetitions	Sustained /a/, three repetitions
Descriptors	44 replicated acoustic descriptors + gender	752 multi-family engineered descriptors + gender
Imbalance handling	Class weighting (balanced folds)	Class weighting (VoxNeuro subspace branch; unaugmented comparators); ordinary SMOTE for SMOTE + logistic regression; G-SMOTE for full-space kernel models
Outer evaluation	Five-fold subject-disjoint cross-validation	

G-SMOTE, Grassmann-geodesic synthetic minority oversampling (Section 2.8); SMOTE, synthetic minority oversampling technique.

2.2. Subject-Disjoint Problem Formulation

Let $R_i=\{(x_{ij},y_i):j=1,\ldots,m_i\}$ denote all recordings from subject i; Table 2 summarizes the notation used throughout. For outer fold k, let I_k^{tr} and I_k^{te} be the training and held-out subject-index sets. Leakage-safe evaluation requires

$$I_k^{\mathrm{tr}}\cap I_k^{\mathrm{te}}=\varnothing,\, I_k^{\mathrm{tr}}\cup I_k^{\mathrm{te}}=\{1,\ldots,n\}. \tag{1}$$

All recordings in R_i are assigned to the same partition as subject i (Figure 2A,B). For a pointwise classification loss ℓ and the model f_k fitted on the training subjects of fold k, the fold risk is evaluated once per held-out subject,

$$\hat{R}_k(f_k)=\frac{1}{|I_k^{\mathrm{te}}|}\sum_{i\in I_k^{\mathrm{te}}}\ell(f_k(Z_i),y_i), \tag{2}$$

where Z_i is the complete subject-level representation. Equations (1) and (2) define the intended estimand, generalization to an unseen person rather than to an unseen row from a known person; the reported balanced accuracy and macro-F1 are fold-level aggregate metrics defined in Equation (17).

Table 2. Principal notation.

Symbol	Meaning
$x_{ij}\in\mathbb{R}^d$	feature vector of recording j of subject i
R_i, y_i	all recordings of subject i and its diagnostic label
$I_k^{\mathrm{tr}}, I_k^{\mathrm{te}}$	training and held-out subject-index sets of outer fold k
M_i	normalized $d\times 3$ recording matrix of subject i (full space)
z_{ij}, q	recording j of subject i projected onto the supervised subspace and the subspace dimension
$M_i^{(q)}$, $s_i^{(q)}$	projected recording matrix ($q\times 3$) and summary ($\mathbb{R}^{2q}$) in the supervised subspace
Q_i, $[Q_i]$	orthonormal basis from the thin SVD and its subspace class
$\mathrm{Gr}(r,d)$, $\mathrm{St}(r,d)$	Grassmann and Stiefel manifolds of rank r in $\mathbb{R}^d$
$P_i=Q_iQ_i^T$	orthogonal projector representing $[Q_i]$

Symbol	Meaning
$\theta_{ij,h}$, d_{ch}	principal angles and chordal distance between subspaces
$s_i \in \mathbb{R}^{2d}$	Euclidean mean–dispersion summary of subject i
K_G, K_E, K; w	Grassmann, Euclidean, and fused kernels (Ψ: direct-sum feature map); fusion weight
γ_G, γ_E	Gaussian kernel coefficients (inverse squared bandwidths) of the two views
$Q_{\mathrm{syn}}(t), s_{\mathrm{syn}}(t)$	geodesic and Euclidean interpolants of G-SMOTE
C_{y_i}; N_k, $N_{k,y}$	class-weighted SVM box constraint; training representations in fold k after augmentation and their class counts
F, $\overline{m}$, SE	number of outer folds, fold-mean metric, standard error
$\overline{I}_{gk}$, π_g, B	fold-level permutation importance (mean over B draws) and permutation of family g
h, A, $R_{\mathrm{seen}}, R_{\mathrm{new}}, R_{\mathrm{row}}$	record-level classifier, leakage event, and associated risks
$\widehat{F}_k^{(l)}$, Φ^{-1}	training-fold empirical distribution function of coordinate l; standard normal quantile function
W_q, τ_{ij}, Q	PLS rotation to q dimensions, class-balanced recording targets, set of subspace dimensions
f_{AS}, $\hat{\sigma}_{k,q}$, n_k^{rec}	adaptive-subspace decision function, training-fold decision scale of member q, number of training recordings

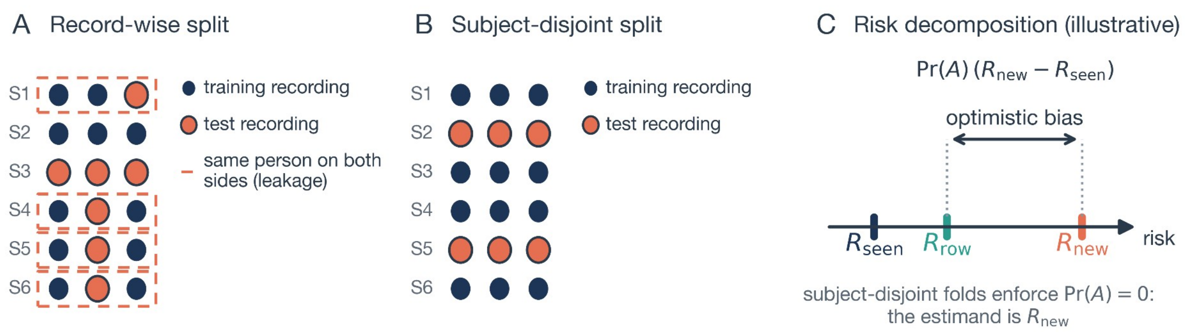

Figure 2. Schematic illustration of evaluation leakage with repeated recordings. (**A**) A record-wise split assigns individual recordings to training or test sets, so several subjects (dashed outlines) appear on both sides and the test set is not independent of training identities. (**B**) A subject-disjoint split assigns all recordings of a subject jointly. (**C**) Consequence of Equation (20): the record-wise estimate R_{row} interpolates between the easier seen-person risk R_{seen} and the deployment-relevant unseen-person risk R_{new}, drawn for an illustrative $\Pr(A) = 0.65$; the arrow marks the optimistic bias of magnitude $\Pr(A)(R_{\mathrm{new}} - R_{\mathrm{seen}})$.

2.3. Normalization and Subject Representation

For subject i, the three acoustic vectors $x_{ij} \in \mathbb{R}^d$ are normalized with functions estimated exclusively from the recordings of training subjects in outer fold k. VoxNeuro uses Gaussian rank normalization. For coordinate l, let $\widehat{F}_k^{(l)}$ denote the interpolated training-quantile transformation of the training-fold recordings, which approximates the empirical distribution function through $J_k = \min(300, n_k^{\mathrm{rec}})$ quantile knots with linear interpolation between them (tied values share one image) ($J_k = 192$ on UCI-489 and 300 on PD-252, where n_k^{rec} is the number of training recordings), and let Φ^{-1} denote the standard normal quantile function, with its argument clipped to $[10^{-7}, 1-10^{-7}]$:

$$\tilde{x}_{ij}^{(l)} = \Phi^{-1}\left(\widehat{F}_k^{(l)}\left(x_{ij}^{(l)}\right)\right), \quad l = 1, \ldots, d, \tag{3}$$

The transformation is monotone in every coordinate and is fitted on training recordings only; held-out recordings are mapped through the frozen training-fold functions. Many engineered acoustic descriptors are heavy-tailed, and under z-standardization a few extreme coordinates dominate the squared Euclidean distances that enter the kernels and the median bandwidth rule, whereas rank normalization reduces the influence of extreme values in continuous descriptors without any use of labels. Tied and discrete predictors do not acquire the same Gaussian marginal: the dataset-provided binary gender covariate is mapped by the same rule to two values near ±5.2, and no subsequent variance normalization is applied, so this coordinate carries a larger weight in the Euclidean view than a continuous coordinate. Z-standardization with a training-fold mean and scale vector (zero-variance coordinates assigned unit scale) is retained as an ablation (Section 2.11). In both cohorts every subject has $m_i = 3$ recordings, so the normalized recordings are stacked column-wise into a $d \times 3$ matrix, and a thin singular value decomposition (SVD) [57] gives

$$\begin{aligned} M_i &= [\tilde{x}_{i1} \tilde{x}_{i2} \tilde{x}_{i3}] = U_i \Sigma_i V_i^{\top}, Q_i = U_i(:, 1:r), \\ Q_i^{\top} Q_i &= I_r, [Q_i] \in \mathrm{Gr}(r, d). \end{aligned} \tag{4}$$

Throughout, the rank is fixed at $r = 3$, matching the three repeated recordings per subject. The basis is not unique: Q_i and $Q_i O$, with O in the orthogonal group $\mathrm{O}(r)$, represent the same subspace. This quotient structure removes coordinate choices within the retained SVD basis (Figure 3A). In the official releases, one subject per cohort (CONT-36 in UCI-489; subject 37 in PD-252) contains byte-identical repeated recordings, so its matrix has numerical rank two. For such a subject, Equation (4) retains the orthonormal completion supplied by the SVD as the third basis vector; that direction is a solver-dependent completion that is not identified by the recordings, so the representation is nominally rank three for these two subjects, and the reference implementation accepts them only under an explicit opt-in flag. In the high-dimensional regime defined in Section 2.7, the same construction is applied to supervised low-dimensional projections of the recordings.

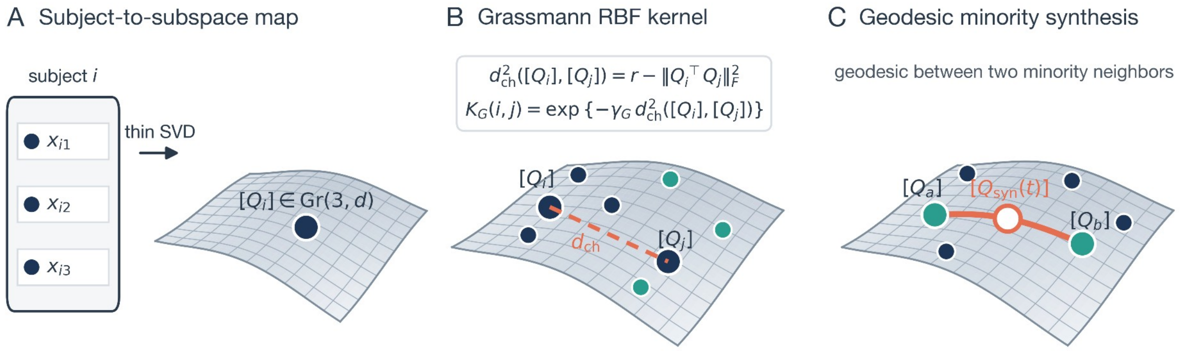

Figure 3. Geometric interpretation of the VoxNeuro representation (schematic; the curved sheet stands for the Grassmann manifold). (**A**) The three repeated recordings of a subject are mapped by a thin SVD to a single point $[Q_i]$ on Gr(3,d). (**B**) The chordal distance d_{ch} between two subject subspaces (dashed chord), whose square enters the displayed Gaussian Grassmann radial-basis-function kernel, compares subjects independently of the arbitrary orthonormal basis selected for each subspace; navy and teal points denote subjects of the two classes. (**C**) Schematic of minority synthesis along the Grassmann geodesic joining two minority neighbors, rather than row-wise Euclidean interpolation.

2.4. Principal Angles and Projection Geometry

Principal angles between two subject subspaces (Figure 4A) are defined by [32,58]

$$Q_i^{\top} Q_j = A_{ij} \cos(\Theta_{ij}) B_{ij}^{\top}, \quad \Theta_{ij} = \mathrm{diag}(\theta_{ij,1}, \ldots, \theta_{ij,r}), \tag{5}$$

where A_{ij} and B_{ij} are orthogonal. With the projector $P_i = Q_i Q_i^{\top}$, the squared chordal distance is given by [33,59]

$$\begin{aligned} d^2_{\mathrm{ch}}([Q_i], [Q_j]) &= ½ \|P_i - P_j\|_F^2 \\ &= r - \|Q_i^{\top} Q_j\|_F^2 = \sum_{h=1}^{r} \sin^2 \theta_{ij,h}. \end{aligned} \tag{6}$$

Because $(Q_i O)(Q_i O)^{\top} = P_i$, the distance is basis invariant (Figure 4B). Vectorizing P_i maps the chordal distance to a Euclidean distance in projector space, up to a constant factor $\sqrt{2}$ absorbed by the kernel coefficient, thereby providing a Euclidean embedding for the Gaussian Grassmann kernel.

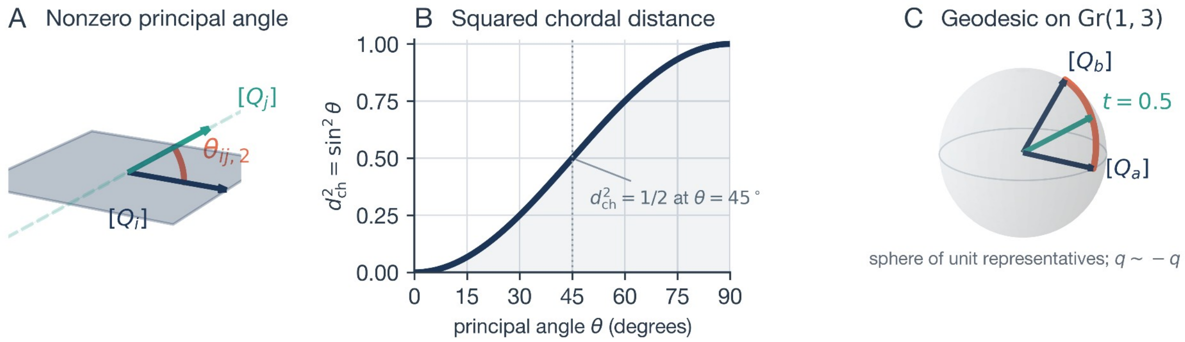

Figure 4. The geometry behind Equations (5), (6), (13), and (14), drawn for illustrative toy subspaces. (**A**) Two planes in $\mathbb{R}^3$ that share one direction, so the first principal angle is zero and the drawn angle is the second principal angle $\theta_{ij,2}$. (**B**) Exact rank-one relation between a principal angle and the squared chordal distance, $d^2_{\mathrm{ch}} = \sin^2\theta$. (**C**) A geodesic path computed with Equations (13) and (14) between two one-dimensional subspaces $[Q_a]$ and $[Q_b]$; every interpolated frame in the drawn path satisfies $Q_{\mathrm{syn}}(t)^\top Q_{\mathrm{syn}}(t) = I_r$ to machine precision, consistent with Proposition 2.

2.5. Complementary Euclidean View

The complementary Euclidean view retains the coordinate-wise mean and dispersion:

$$
\begin{aligned}
\mu_i &= \tfrac{1}{3}\sum_{j=1}^{3} \tilde{x}_{ij}, \sigma_i = \Big[\tfrac{1}{3}\sum_{j=1}^{3} (\tilde{x}_{ij} - \mu_i)^{\oslash 2}\Big]^{1/2}, \\
s_i &= [\mu_i^\top \sigma_i^\top]^\top \in \mathbb{R}^{2d},
\end{aligned}
\tag{7}
$$

where $(\cdot)^{\oslash 2}$ and the square root are applied elementwise. Thus $[Q_i]$ encodes the orientation of the span of the normalized repetitions (M_i is not centered within subject, so this span generally includes the mean direction), whereas s_i retains coordinate-level location and dispersion.

2.6. Direct-Sum Kernel and Guarantee of Positive Semidefiniteness

Let ϕ_G and ϕ_E be feature maps for Gaussian kernels on the projection and Euclidean views [38,39]. With fusion weight w, the joint representation is the Hilbert direct sum

$$
\begin{aligned}
\Psi(Z_i) &= \left(\sqrt{w}\,\phi_G([Q_i]), \sqrt{1-w}\,\phi_E(s_i)\right), \\
&\qquad 0 \le w \le 1,
\end{aligned}
\tag{8}
$$

which induces, for kernel coefficients $\gamma_G, \gamma_E > 0$ (inverse squared bandwidths),

$$
\begin{aligned}
K_G(i,j) &= \exp\{-\gamma_G d^2_{\mathrm{ch}}([Q_i],[Q_j])\}, \\
K_E(i,j) &= \exp\{-\gamma_E \|s_i - s_j\|_2^2\}, \\
K(i,j) &= w K_G(i,j) + (1-w) K_E(i,j).
\end{aligned}
\tag{9}
$$

Proposition 1 *(basis invariance and positive semidefiniteness).* The kernel in Equation (9) is invariant to $Q_i \mapsto Q_i O$ and is PSD. Basis invariance follows from the projector representation in Equation (6). Each Gaussian block is PSD on its Euclidean embedding, and a nonnegative weighted sum of PSD kernels is PSD [39,41]; hence, for every $c \in \mathbb{R}^n$,

$$
c^\top K c = w\, c^\top K_G c + (1-w) c^\top K_E c \ge 0. \tag{10}
$$

Equivalently, $K(i,j) = \langle \Psi(Z_i), \Psi(Z_j) \rangle$ in the direct-sum reproducing-kernel Hilbert space. □ The projector embedding is essential: Gaussian kernels built on other manifold distances, such as the geodesic arc length, are not PSD for all bandwidths [37], whereas the chordal construction used here inherits positive semidefiniteness from its explicit Euclidean embedding. The invariance applies to alternative bases of the same retained subspace; for the two rank-deficient subject matrices of Section 2.3, the third completion direction is not identified by the data, so distances involving that direction may depend on the numerical SVD implementation even though they are invariant to the basis chosen for a given completion.

2.7. Dimension-Adaptive Supervised Subspace

Rank normalization leaves a second difficulty untouched: distance concentration. When the feature count d is large relative to the number of recordings, the rank-three subspaces of different subjects become nearly orthogonal, and the squared chordal distances of Equation (6) crowd toward their maximum value r; a Gaussian kernel on nearly constant distances carries little information. On PD-252 the coefficient of variation of the pairwise chordal distance between training subjects (means over the five reference-partition training folds) was 0.019 under z-standardization and 0.035 under rank normalization, compared with 0.052 and 0.066 on the 45-dimensional UCI-489 cohort. VoxNeuro therefore applies a supervised low-dimensional projection fitted on the training-fold recordings: a partial least squares (PLS) regression [60,61] of the normalized recordings on class-count-coded binary targets $\tau_{ij}=y_i/n^{\mathrm{rec}}_{k,y_i}$, where $y_i\in\{-1,+1\}$ is the label of subject i, carried by each of its recordings, and $n^{\mathrm{rec}}_{k,y}$ is the number of training recordings with label y. For this single-response regression the coding is an affine transformation of the binary labels, so it does not implement sample-weighted PLS; class imbalance is handled downstream by the SVM class weights. The regression was fitted without target or predictor scaling; with $W_q\in\mathbb{R}^{d\times q}$ the fitted rotation matrix and $\bar{\tilde{x}}_k$ the mean of the normalized training recordings, every recording is projected as

$$\begin{aligned} z_{ij} &= W_q^{\top}(\tilde{x}_{ij}-\bar{\tilde{x}}_k)\in\mathbb{R}^q,\ \tau_{ij}=\frac{y_i}{n^{\mathrm{rec}}_{k,y_i}}, \\ M_i^{(q)} &= [z_{i1}\,z_{i2}\,z_{i3}],[Q_i^{(q)}]\in\mathrm{Gr}(3,q),s_i^{(q)}\in\mathbb{R}^{2q}, \end{aligned} \tag{11}$$

and the subject views of Sections 2.3–2.5, the median bandwidth rule, and the fused kernel of Equation (9) are then built in $\mathbb{R}^q$ exactly as in the full space, with the shared weight $w=0.5$. Because the projection is supervised, it is refitted inside every training fold and never sees a held-out recording. After projection to $q=16$ dimensions the coefficient of variation of the PD-252 chordal distances rose to 0.064. One class-weighted SVM f_q is trained for each dimension $q\in Q=\{8,16,32\}$. Instead of selecting a dimension within the fold, the decision values of the three members are divided by their training-fold standard deviations $\hat{\sigma}_{k,q}$ (population standard deviation of the members' decision values over the training subjects, whose mean is $\bar{f}_{k,q}$, plus 10^{-12}) and averaged, giving a fixed-dimension ensemble that performs no within-fold dimension selection:

$$\begin{aligned} f_{\mathrm{AS}}(Z) &= \frac{1}{|Q|}\sum_{q\in Q}\frac{f_q(Z)}{\hat{\sigma}_{k,q}},\hat{\sigma}_{k,q}=\Big[\frac{1}{|I_k^{\mathrm{tr}}|}\sum_{i\in I_k^{\mathrm{tr}}}(f_q(Z_i)-\bar{f}_{k,q})^2\Big]^{1/2}+10^{-12}, \\ \text{activated iff}\quad d &> n_k^{\mathrm{rec}}=\sum_{i\in I_k^{\mathrm{tr}}} m_i. \end{aligned} \tag{12}$$

The stage is activated by a fixed dimensionality gate: it is applied only when d exceeds the number of training recordings n_k^{rec}, which holds on PD-252 ($d=753$, $n_k^{\mathrm{rec}}=603$ or 606) and not on UCI-489 ($d=45$, $n_k^{\mathrm{rec}}=192$). The gate, the dimension set, and the equal-weight average are part of the shared configuration of Section 2.11. When the gate is inactive the model reduces to the full-space fused model of Equation (9) under rank normalization, including the conditional G-SMOTE branch of Section 2.8 for imbalanced training folds. Inside the supervised subspace no synthetic subjects are generated; the class-weighted SVM of Section 2.9 balances the classes, and the corresponding oversampling ablation is reported in Section 3.3. The complete model is referred to as VoxNeuro. Its ablations are the rank-normalized full-space model (no subspace stage), the z-standardized full-space model, the full-space Euclidean-kernel ablation ($w=0$), and the subspace-ensemble $w=0$ and $w=1$ variants.

2.8. Grassmann-Geodesic Synthetic Minority Oversampling

In the full-space branch, imbalanced training folds are balanced by a project-specific manifold adaptation of synthetic oversampling, referred to as Grassmann-geodesic SMOTE (G-SMOTE); the abbreviation does not denote the separate Geometric SMOTE algorithm of Euclidean imbalanced learning [62]. Following the synthetic-neighbor principle of standard SMOTE [44] and its descendants [45,46,63], for minority subjects a and b,

$$\begin{aligned} Q_a^{\top} Q_b &= U \cos(\Theta) V^{\top}, \\ W_{ab} &= (Q_b V - Q_a U \cos\Theta)(\sin\Theta)^{\dagger}, \end{aligned} \tag{13}$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse. A shortest geodesic and its coupled Euclidean interpolation are defined as follows (Figure 4C):

$$\begin{aligned} Q_{\mathrm{syn}}(t) &= [Q_a U \cos(t\Theta) + W_{ab} \sin(t\Theta)] V^{\top}, \\ s_{\mathrm{syn}}(t) &= s_a + t(s_b - s_a), \quad t \sim U(0,1). \end{aligned} \tag{14}$$

Within each imbalanced training fold, the minority class was the control class. For each synthetic subject, one minority subject was sampled uniformly at random, and a partner was selected uniformly from that subject's five nearest minority neighbors under squared chordal distance. Synthetic subjects were generated until the two class counts were equal. The same sampled value of t was used for the Grassmann and Euclidean interpolations. For numerical stabilization, singular values were clipped to $[-1,1]$ before the arccosine, directions with $\sin\theta \le 10^{-10}$ were treated as zero-angle directions, and the implementation applied a QR factorization to $Q_{\mathrm{syn}}(t)$. The resulting invertible right basis change preserves the represented Grassmann point. No synthetic subjects were generated for balanced UCI-489. Interpolating whole-subject representations along the manifold avoids creating synthetic recording rows and selects partners by subspace geometry, but it still interpolates high-dimensional summaries and therefore remains susceptible to the noise-amplification risks of oversampling [47].

Proposition 2 *(manifold validity).* In exact arithmetic, the nonzero-angle columns of $Q_a U$ and W_{ab} are mutually orthonormal, while zero-angle directions remain in the cosine term. Consequently,

$$Q_{\mathrm{syn}}(t)^{\top} Q_{\mathrm{syn}}(t) = V[\cos^2(t\Theta) + \sin^2(t\Theta)] V^{\top} = I_r. \tag{15}$$

Therefore, $Q_{\mathrm{syn}}(t) \in \mathrm{St}(r,d)$, the Stiefel manifold of orthonormal $d \times r$ frames; it represents a valid Grassmann point. Its endpoints are equivalent to $[Q_a]$ and $[Q_b]$. □ Proposition 2 establishes validity of the Grassmann component only; the coupled synthetic representation pair is not guaranteed to correspond to a realizable set of three acoustic recordings.

2.9. Kernel Support Vector Machine, Metrics, and Complexity

For support vector machine (SVM) training [64], labels are recoded as $y_i \in \{-1,+1\}$, $Y_k = \mathrm{diag}(y_1, \dots, y_{N_k})$ over the training representations of fold k, and balanced class weighting sets $C_{y_i} = C N_k / (2 N_{k,y_i})$, where N_k is the number of training representations in outer fold k after any training-only augmentation and $N_{k,y}$ is the corresponding class count; this is the balanced heuristic implemented in standard open-source libraries [65]. When the training classes are exactly balanced, as on UCI-489 and after G-SMOTE parity on PD-252, these weights reduce to $C_{y_i} = C$; the weighting is therefore active for the supervised-subspace branch of VoxNeuro, which generates no synthetic subjects, and for the unaugmented comparators on imbalanced PD-252. The class-weighted dual and the decision function with bias b are

$$\begin{aligned} \max_{\alpha} \quad & 1^{\top}\alpha - ½\alpha^{\top} Y_k K_k Y_k \alpha, \\ \text{subject to} \quad & 0 \le \alpha_i \le C_{y_i}, \; y^{\top}\alpha = 0, \quad i = 1, \dots, N_k, \\ f(Z) \quad & = \sum_{i=1}^{N_k} \alpha_i y_i K(Z_i, Z) + b. \end{aligned} \tag{16}$$

Balanced accuracy (BA) [66] and macro-F1 [67] weight diagnostic classes symmetrically:

$$\begin{aligned} \mathrm{BA} &= ½(\mathrm{TPR} + \mathrm{TNR}), \\ \text{macro-F1} &= ½(F1_{\mathrm{PD}} + F1_{\mathrm{Ctrl}}), \end{aligned} \tag{17}$$

where TPR is sensitivity, TNR is specificity, and $F1_{\mathrm{PD}}$ and $F1_{\mathrm{Ctrl}}$ are the class-wise F1 scores. Balanced accuracy averages the two class recalls, whereas macro-F1 averages the two class-specific F1 scores; both assign equal weight to the diagnostic classes, and the Matthews correlation coefficient is a common alternative [68]. For $F = 5$ outer folds, a metric m is summarized by

$$\begin{aligned} \overline{m} &= \frac{1}{F}\sum_{k=1}^{F} m_k, \\ \mathrm{SE}(\overline{m}) &= \left[\frac{1}{F(F-1)}\sum_{k=1}^{F}(m_k-\overline{m})^2\right]^{1/2}. \end{aligned} \tag{18}$$

The reported standard error (SE) is a descriptive measure of fold-to-fold variability and is not an inferential confidence interval: because the training sets of the folds overlap, fold scores are dependent, and this quantity is a descriptive rescaling of fold variability rather than an estimate of independent-replication uncertainty.

For n subjects and rank $r=3$, subject SVDs require $O(ndr^2)$, the Grassmann Gram matrix requires $O(n^2dr^2)$, and the Euclidean block requires $O(n^2d)$. These bounds cover view construction only and exclude the kernel-SVM optimization, which can dominate as n grows; the subspace branch additionally requires three PLS fits and projections, three kernel constructions, and three kernel-SVM optimizations, and the full-space branch adds the G-SMOTE neighbor search where active. All components remain tractable for the present cohort sizes $n=80$ and $n=252$.

2.10. Formal Effect of Record-Wise Leakage

The mechanism illustrated in Figure 2 can be stated formally, following record-wise versus subject-wise analyses in clinical machine learning [18,20,21]. Let (X,y) be a random validation recording and its label, let A be the event that the recording belongs to a person represented in training, and let h be a record-level classifier. With A^c the complement of A, define

$$\begin{aligned} R_{\text{seen}} &= \mathrm{E}[\ell(h(X),y)\mid A], \\ R_{\text{new}} &= \mathrm{E}[\ell(h(X),y)\mid A^c]. \end{aligned} \tag{19}$$

A record-wise split then estimates

$$\begin{aligned} R_{\text{row}} &= \Pr(A)R_{\text{seen}} + [1-\Pr(A)]R_{\text{new}}, \\ R_{\text{row}} - R_{\text{new}} &= \Pr(A)(R_{\text{seen}} - R_{\text{new}}). \end{aligned} \tag{20}$$

If repeated recordings from previously observed people are easier to classify than recordings from unseen people, then $R_{\text{seen}} < R_{\text{new}}$, and the record-wise estimate is optimistically biased in direct proportion to $\Pr(A)$ (Figure 2C). Subject-disjoint folds enforce $\Pr(A)=0$ by construction.

2.11. Experimental Protocol, Comparators, and Ablations

Outer folds were generated by five-fold stratified subject-level cross-validation with shuffling and a random seed of 42, referred to as the reference partition (Figure 5A). For each outer fold, the recording-level normalizer was fitted only to recordings from training subjects, and each kernel coefficient was estimated from the positive squared pairwise distances δ^2 of the corresponding view over the training subjects (after G-SMOTE augmentation where active) as $\gamma = [2\ \text{median}(\delta^2) + 10^{-12}]^{-1}$. A single final configuration, $r=3$, $w=0.5$, $C=1$, and $Q=\{8,16,32\}$, was applied to both cohorts. This configuration was chosen after inspection of reference-partition results from both cohorts; consequently, the reference outer folds participated indirectly in method development, and the corresponding estimates are design-stage rather than unbiased validation estimates [23,24]. Within each training fold of the full-space branch, resampling was governed solely by the observed class counts: if the counts were unequal, the minority class was raised to the majority count using G-SMOTE for the full-space kernel models and ordinary SMOTE only for the comparator explicitly identified as SMOTE-based; if the counts were equal, resampling was the identity operation. All UCI-489 training folds were balanced, so no synthetic subjects were generated there; on PD-252 the full-space comparators activated the branch in every fold, whereas VoxNeuro, whose gate activates the supervised-subspace branch on PD-252, generated no synthetic subjects on either cohort and is therefore deterministic given the outer partition. Table 3 summarizes the shared configuration, and the implementation is available in the project repository (see the Data Availability Statement).

VoxNeuro was compared with five standard comparators trained on the same subject-level summary s_i under the same training-fold rank normalization and identical folds: class-weighted logistic regression; a class-weighted linear SVM; SMOTE followed by logistic regression where resampling is active; a class-weighted random forest (500 trees, ordinary bootstrap sampling, square-root feature subsampling, unrestricted depth, class weights recomputed within each bootstrap sample); and partial least squares discriminant analysis (PLS-DA). Logistic regression used L2 regularization with $C=1$ (liblinear solver, at most 5000 iterations); the linear SVM used the L2 penalty with squared-hinge loss and $C=1$ (at most 20,000 iterations); SMOTE used five neighbors, or one fewer than the minority count when necessary; a subject was labeled PD when its decision score was strictly positive, and a forest probability of exactly 0.5 was assigned to control. For PLS-DA, the subject summaries were standardized with the training subjects, projected by PLS regression on the same class-count-coded targets without scaling to q components, and classified by shrinkage linear discriminant analysis with equal priors; $q \in \{2,4,8,16\}$ was chosen by pooled inner five-fold out-of-fold balanced accuracy inside each training fold, with the smallest q retained on ties, and the scaler, projection, and discriminant were then refitted on the complete training fold. Inner selection was conditional on the recording-level normalizer fitted on the complete outer-training fold: the summary scaler, projection, and discriminant were refitted within each inner split, whereas the recording-level normalizer was not. A full-space Euclidean-kernel ablation with fusion weight $w=0$ shares the summary features, the resampling, and the kernel family of the full-space model and differs from it only in the Grassmann kernel term (Figure 5B); on PD-252 it retains the G-SMOTE branch and is therefore reported as an ablation rather than as a standard comparator. All comparators were implemented in the released codebase and verified against their stated configurations; the term audited comparators refers to this set, as distinct from values reported by other studies. Three groups of ablations were evaluated: (i) normalization, comparing the rank-normalized and z-standardized full-space models; (ii) the subspace stage, removing it (rank-normalized full-space model) and, with the gate forced on for UCI-489, applying it where the rule deactivates it, together with single members $q \in \{4,8,16,32,64\}$ against the three-member ensemble and geodesic oversampling inside the subspace branch across 24 augmentation seeds; and (iii) the views, setting $w=0$ (Euclidean only) and $w=1$ (Grassmann only) inside the subspace ensemble (subspace-ensemble $w=0$ and $w=1$ ablations). Supplementary analyses of a z-standardized full-space variant (augmentation-seed sensitivity, rank $r=2$, augmentation family, fusion-weight sweep, and feature-family permutation sensitivity) are reported in Appendix A.

To assess sensitivity to the partition, the entire five-fold evaluation was repeated for 20 alternative stratified subject partitions (StratifiedKFold seeds 0–19) with every model, and paired differences were summarized across partitions. For models with the G-SMOTE branch the augmentation seed is derived from the partition seed, and for the other stochastic comparators the estimator or resampling random state also changes with the partition seed, so their summaries combine partition sensitivity with algorithmic randomness; VoxNeuro itself is deterministic conditional on the partition, and the archived fixed-augmentation partition results of the z-standardized full-space variant are used as a check. These results are an internal resampling-sensitivity analysis on the same 80 and 252 subjects: they do not constitute independent validation, partition-level counts are descriptive, and the reuse of the same subjects for design and sensitivity analysis leaves an analysis-level selection optimism that only a new cohort can remove. All analyses used Python 3.11.15 with NumPy 1.26.4, SciPy 1.11.4, pandas 2.0.3, scikit-learn 1.3.2, imbalanced-learn 0.11.0, and threadpoolctl 3.6.0 under single-threaded linear algebra.

Table 3. Final shared analysis configuration. All learned components are fitted inside the outer training folds only.

Component	Setting
Outer evaluation	five-fold stratified subject-level cross-validation, shuffled, seed 42
Normalization	Gaussian rank normalization fitted on training recordings ($\min(300, n_k^{\mathrm{rec}})$ quantile knots); z-standardization (training-fold mean / scale, unit scale for zero-variance coordinates) as an ablation

Component	Setting
Supervised subspace	PLS projection fitted on training recordings with class-balanced targets; members $q \in \{8, 16, 32\}$ averaged after division by the training-fold decision SD; activated only when d exceeds the number of training recordings (PD-252: 753 > 603 or 606; UCI-489: 45 < 192); no synthetic subjects in this branch
Subject representation	rank $r = 3$ thin SVD (Grassmann view) + mean/dispersion vector (Euclidean view)
Kernel coefficients	$\gamma = [2 \ \mathrm{median}(\delta^2) + 10^{-12}]^{-1}$ from positive squared pairwise training distances (augmented training set in the full-space branch on PD-252)
Fusion weight / penalty	$w = 0.5$; class-weighted SVM with $C = 1$; one shared post-development setting for both cohorts
Resampling rule	training-fold class-count gate in the full-space branch: identity when balanced (all UCI-489 folds); G-SMOTE to class parity when imbalanced, five nearest minority neighbors under chordal distance; the supervised-subspace branch uses class weighting only
Comparators	standard comparators: PLS-DA (components by inner cross-validation), class-weighted random forest (500 trees), class-weighted logistic regression, class-weighted linear SVM, SMOTE + logistic regression where resampling is active; full-space Euclidean-kernel ablation (fusion weight 0); all under rank normalization on identical folds
Metrics	balanced accuracy and macro-F1, mean ± fold-level SE; pooled out-of-fold confusion counts
Ablations and sensitivity	z-standardization; no subspace stage; $w = 0$ and $w = 1$ view ablations with the automatic gate (subspace ensemble on PD-252, rank-normalized full-space model on UCI-489); single subspace dimensions 4–64; gate forced on (UCI-489); G-SMOTE inside the subspace branch (24 seeds); 20 outer subject partitions for every model; full-space branch: 24 augmentation seeds, rank $r = 2$, augmentation family, fusion-weight sweep, feature-family permutation (Appendix A)

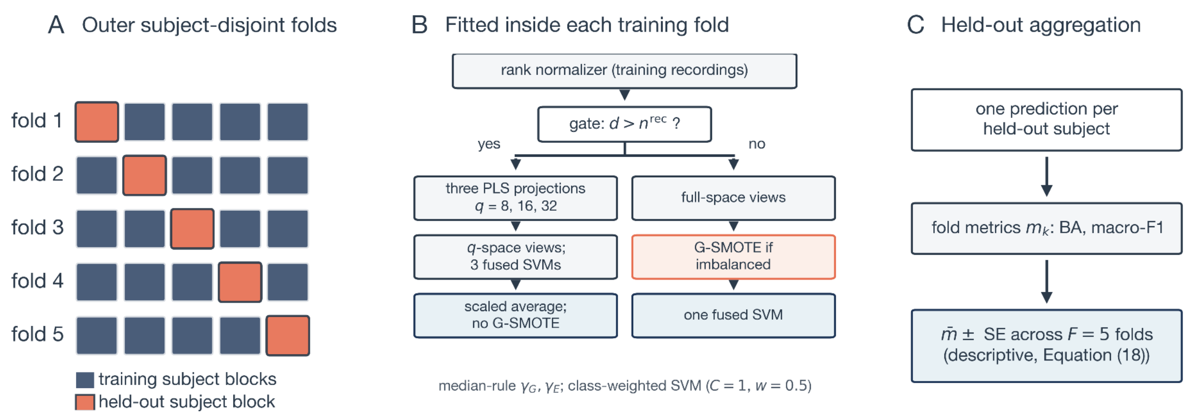

Figure 5. Evaluation protocol. (**A**) Five-fold subject-disjoint partition; in each outer fold one subject block (orange) is held out and never influences any fitted component. (**B**) Everything that learns from data is fitted inside the training fold only: the rank normalizer, then either the supervised-subspace branch (dimensionality gate active: PLS projections, q-space views, three SVMs averaged after scaling; no synthetic subjects) or the full-space branch (gate inactive: full-space views, conditional G-SMOTE in imbalanced folds, one SVM), with median-rule kernel coefficients in both branches. (**C**) Exactly one prediction is issued per held-out subject, fold metrics are computed at the subject level, and Equation (18) aggregates them descriptively.

3. Results

3.1. Performance at the Reference Partition

Figure 6 and Table 4 report VoxNeuro, its ablations, and the standard comparators at the reference partition. On UCI-489, VoxNeuro reached a balanced accuracy of 0.875 (SE 0.040) and a macro-F1 of 0.873 (SE 0.040), compared with 0.863 and 0.860 for the full-space Euclidean-kernel ablation, 0.825 and 0.822 for PLS-DA, 0.800 and 0.799 for the class-weighted random forest, 0.800 and 0.796 for class-weighted logistic regression, and 0.800 and 0.797 for the class-weighted linear SVM; the z-standardized full-space model reached 0.850 and 0.847. On PD-252, VoxNeuro reached 0.816 (SE 0.040) and 0.811 (SE 0.034); the standard comparators reached 0.777 and 0.786 (SMOTE plus logistic regression), 0.769 and 0.774 (class-weighted logistic regression), 0.747 and 0.744 (PLS-DA), 0.734 and 0.747 (class-weighted linear SVM), and 0.671 and 0.696 (class-weighted random forest), and the full-space Euclidean-kernel ablation 0.766 and 0.757. The ablations locate the difference: the rank-normalized full-space model reached 0.782 and 0.781 and the z-standardized full-space model 0.731 and 0.749, so rank normalization was associated with +0.051 balanced accuracy and +0.032 macro-F1 (rank-normalized versus z-standardized full-space model) and replacing the rank-normalized full-space model (with G-SMOTE) by the unaugmented supervised-subspace ensemble with a further +0.034 and +0.029; this second comparison jointly changes projection, ensembling, and oversampling and does not isolate their individual contributions; the three ensemble members individually reached 0.809, 0.814, and 0.798 balanced accuracy for $q = 8$, 16, and 32. Under z-standardization a Grassmann-only kernel was insufficient on its own on both cohorts (balanced accuracy 0.600 on UCI-489 and 0.636 on PD-252; Appendix A). These are descriptive cross-validation estimates at one outer partition and, for VoxNeuro, design-stage estimates (Section 2.11); Section 3.3 reports the partition-sensitivity analysis, and no inferential test of model superiority was performed.

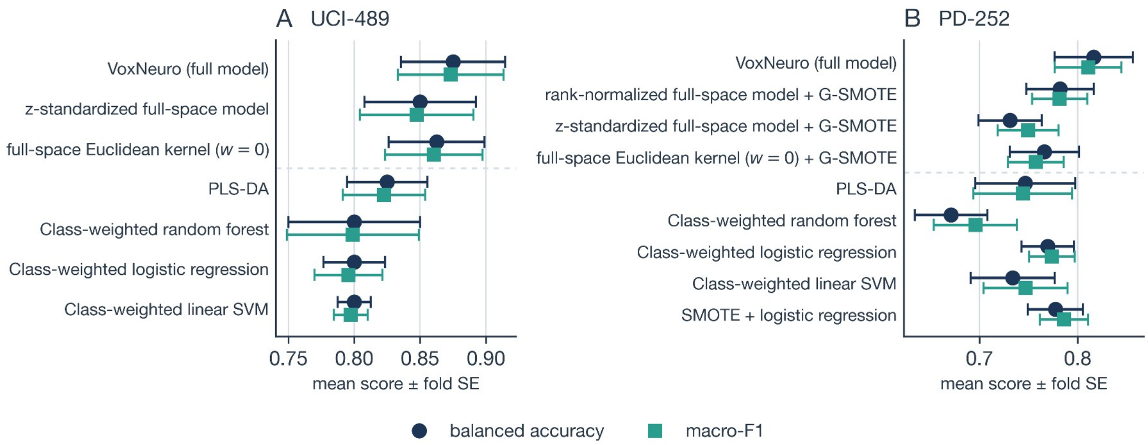

Figure 6. Subject-level scores at the reference partition for (**A**) class-balanced UCI-489 and (**B**) high-dimensional, class-imbalanced PD-252, drawn from the archived out-of-fold results. The full model and its ablations are shown above the dashed separator and the standard comparators below it, all under the same training-fold rank normalization except the z-standardized full-space model. Markers denote mean balanced accuracy (circles) and macro-F1 (squares); whiskers denote the descriptive fold-level standard error across five outer folds, not an inferential interval. On UCI-489 the subspace stage is inactive, so the rank-normalized full-space model coincides with the full model and is not shown, and SMOTE plus logistic regression coincides with logistic regression; on PD-252 the full-space Euclidean-kernel ablation retains the G-SMOTE branch.

Table 4. Subject-level performance at the reference partition (mean ± fold-level standard error over five subject-disjoint folds). All models except the z-standardized full-space model share the same training-fold rank normalizer and outer folds; kernel models use the median-bandwidth rule and $C = 1$, whereas the other comparators retain their model-specific fitting procedures. Rows list the full model, its ablations, and the standard comparators. On UCI-489 the subspace stage is inactive, so the rank-normalized full-space model coincides with the full model, and SMOTE plus logistic regression coincides with logistic regression; on PD-252 the full-space Euclidean-kernel ablation and the z-standardized full-space model use the archived default augmentation seed. The fold-level SE is descriptive and not an inferential interval. Reference-partition estimates are design-stage estimates (Section 2.11); Table 6 reports the partition-sensitivity analysis.

Dataset	Model	Balanced acc.	Macro-F1
UCI-489	VoxNeuro (full model)	0.875 ± 0.040	0.873 ± 0.040
UCI-489	Z-standardized full-space model	0.850 ± 0.042	0.847 ± 0.043
UCI-489	Full-space Euclidean-kernel ablation ($w = 0$)	0.863 ± 0.036	0.860 ± 0.037
UCI-489	PLS-DA	0.825 ± 0.031	0.822 ± 0.031
UCI-489	Class-weighted random forest	0.800 ± 0.050	0.799 ± 0.050
UCI-489	Class-weighted logistic regression	0.800 ± 0.023	0.796 ± 0.026
UCI-489	Class-weighted linear SVM	0.800 ± 0.012	0.797 ± 0.013
PD-252	VoxNeuro (full model)	0.816 ± 0.040	0.811 ± 0.034
PD-252	Rank-normalized full-space model (no subspace stage; G-SMOTE)	0.782 ± 0.035	0.781 ± 0.028
PD-252	Z-standardized full-space model (G-SMOTE)	0.731 ± 0.032	0.749 ± 0.031
PD-252	Full-space Euclidean-kernel ablation ($w = 0$; G-SMOTE)	0.766 ± 0.035	0.757 ± 0.028
PD-252	PLS-DA	0.747 ± 0.051	0.744 ± 0.050
PD-252	Class-weighted random forest	0.671 ± 0.037	0.696 ± 0.042
PD-252	Class-weighted logistic regression	0.769 ± 0.027	0.774 ± 0.023
PD-252	Class-weighted linear SVM	0.734 ± 0.043	0.747 ± 0.043
PD-252	SMOTE + logistic regression	0.777 ± 0.028	0.786 ± 0.025

3.2. Operating Characteristics

On UCI-489, VoxNeuro produced 36 true-negative (control), 4 false-positive, 6 false-negative, and 34 true-positive (PD) classifications, yielding sensitivity 0.850, specificity 0.900, positive predictive value (PPV) 0.895, and negative predictive value (NPV) 0.857. On PD-252 the pooled confusion matrix contained 169 true-positive, 19 false-negative, 47 true-negative, and 17 false-positive classifications, corresponding to sensitivity 0.899, specificity 0.734, PPV 0.909, and NPV 0.712 (Figure 7, Table 5); the z-standardized full-space model at the archived default augmentation seed had sensitivity 0.931 and specificity 0.531, so the full model trades a small loss of sensitivity for a substantially higher specificity. Sensitivity and specificity do not depend algebraically on prevalence but may change under spectrum and setting shifts, whereas PPV and NPV additionally depend directly on this cohort's 188/64 composition and the default decision threshold and do not transport to lower-prevalence screening settings. All values in this section are reference-partition estimates, and rates computed from pooled confusion matrices are descriptive and need not equal the arithmetic mean of fold-level metrics. Even at the improved operating point, roughly one control in four would be referred at the default threshold, which implies a substantial false-referral burden in lower-prevalence screening populations; this operating profile warrants further evaluation of the method as a research-stage screening aid and does not support its use for autonomous diagnosis.

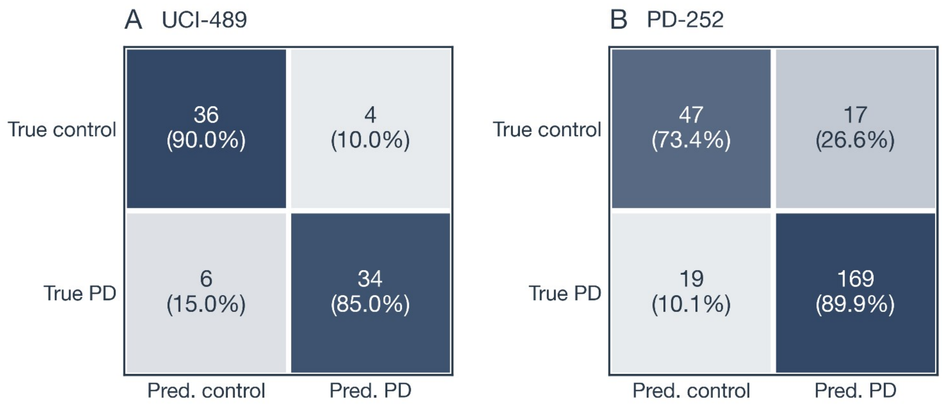

Figure 7. Pooled out-of-fold confusion matrices of VoxNeuro for (**A**) UCI-489 and (**B**) PD-252 at the reference partition, containing one prediction per subject; percentages are normalized within the true class. The model generated no synthetic subjects on either evaluated cohort, so the counts do not depend on an augmentation seed; they are reference-partition estimates and not externally validated operating characteristics.

Table 5. Pooled out-of-fold operating characteristics of VoxNeuro computed from the confusion counts of Figure 7 (UCI-489: TP = 34, FN = 6, TN = 36, FP = 4; PD-252: TP = 169, FN = 19, TN = 47, FP = 17). TP, true positive; FN, false negative; TN, true negative; FP, false positive; PPV, positive predictive value; NPV, negative predictive value.

Dataset	Sensitivity	Specificity	PPV	NPV
UCI-489	0.850	0.900	0.895	0.857
PD-252	0.899	0.734	0.909	0.712

3.3. Partition Sensitivity and Ablations

Figure 8 and Table 6 summarize the partition-sensitivity analysis and the ablations of Section 2.11. Across 20 alternative outer subject partitions on PD-252, VoxNeuro averaged 0.784 ± 0.017 balanced accuracy and 0.785 ± 0.014 macro-F1 (mean ± SD across partitions). It had higher mean scores than every standard comparator. Against class-weighted logistic regression, the highest-scoring standard comparator across both metrics (0.752 ± 0.024 and 0.757 ± 0.024), the paired differences were +0.032 ± 0.020 and +0.028 ± 0.018, positive in 19 of 20 partitions for each metric; against the remaining standard comparators it was higher in at least 19 of 20 partitions on both metrics. Against the full-space Euclidean-kernel ablation (0.752 ± 0.018 and 0.744 ± 0.021) the differences were +0.031 ± 0.015 and +0.041 ± 0.019, positive in all 20 partitions for each metric. Relative to the z-standardized full-space model (0.721 ± 0.020 and 0.734 ± 0.019) the paired differences were +0.062 ± 0.018 and +0.050 ± 0.017, positive in all 20 partitions on both metrics; because the ablation's augmentation seed varies with the partition in these runs, the comparison was repeated against the archived fixed-augmentation partition results of that model, giving +0.059 ± 0.016 and +0.048 ± 0.017, again positive in all 20 partitions. Rank normalization was associated with +0.028 and +0.017 of this margin (rank-normalized versus z-standardized full-space model, positive in 19 of 20 and 17 of 20 partitions) and the supervised-subspace branch with the remainder (VoxNeuro versus the rank-normalized full-space model: +0.034 and +0.033, positive in 19 of 20 and 20 of 20 partitions); the branch comparison bundles projection, ensembling, and the removal of oversampling. On UCI-489, where the subspace stage is inactive, VoxNeuro averaged 0.853 ± 0.019 and 0.852 ± 0.019. It was higher than PLS-DA in 12 of 20 partitions in balanced accuracy and 13 of 20 in macro-F1, but higher than the full-space Euclidean-kernel ablation in only 9 and 10 of 20 (with 2 and 1 ties; mean paired differences +0.0025 and +0.0026), so the UCI-489 margin over the Euclidean view was not partition-consistent. Its mean estimates were similar to those of the z-standardized full-space model (0.853 ± 0.013 and 0.851 ± 0.014; paired balanced-accuracy difference +0.0006 ± 0.0213, positive in 9 of 20 partitions with 4 ties). Forcing the subspace stage on for UCI-489 lowered balanced accuracy by 0.011 on average (higher in only 4 of 20 partitions), which is consistent with the gate rule, although the mechanism was not tested directly.

Figure 8B shows the subspace-dimension sweep with the gate forced on for both cohorts. On PD-252 single members reached 0.769 ($q = 4$), 0.772 ($q = 8$), 0.782 ($q = 16$), 0.782 ($q = 32$), and 0.775 ($q = 64$) mean balanced accuracy across partitions, and the three-member ensemble 0.784, so the fixed-dimension average was at least as good as any single dimension without using held-out subjects to choose one; on UCI-489 every subspace dimension (0.839–0.843) fell below the full-space model (0.853). Inside the subspace branch, geodesic oversampling to class parity changed balanced accuracy by −0.019 ± 0.008 and macro-F1 by −0.007 ± 0.008 across the 24 augmentation seeds (positive in 1 and 7 seeds; Table 6); in this ablation each member's decision scale was computed over its complete augmented training set, so the comparison also changes the distribution used for ensemble scaling. The class-weighted SVM without synthetic subjects was retained for this branch. The view ablations retained the automatic gate, so they evaluate the supervised-subspace ensemble on PD-252 and the rank-normalized full-space model on UCI-489. Across partitions, the Grassmann-only variants ($w = 1$) reached balanced accuracies of 0.636 and 0.579, respectively, and the Euclidean-only variants ($w = 0$) reached 0.779 and 0.851. Fusion at $w = 0.5$ raised mean balanced accuracy by +0.0048 on PD-252 (positive in 12 of 20 partitions) and by +0.0025 on UCI-489 (9 of 20, 2 ties). The Euclidean view therefore carried most of the predictive performance; adding the Grassmann term produced small, partition-inconsistent mean increments, and the Grassmann-only variants performed poorly.

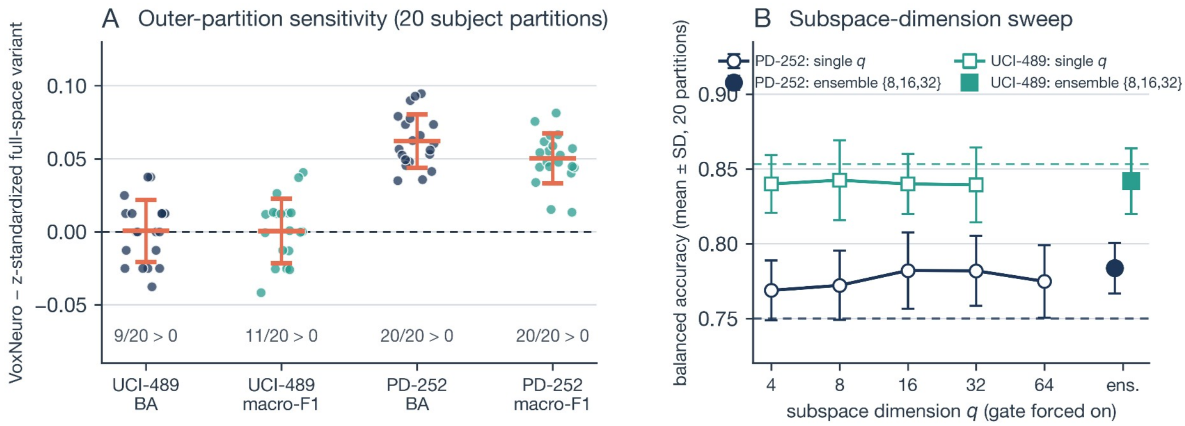

Figure 8. Sensitivity of VoxNeuro to the outer subject partition and to the subspace dimension. (**A**) Paired differences between VoxNeuro and the z-standardized full-space model across 20 alternative outer subject partitions (points; coral bars denote mean ± SD; counts give the number of partitions with a positive difference), for balanced accuracy (BA, navy) and macro-F1 (teal) on both cohorts; the ablation's augmentation seed varies jointly with the partition, and on UCI-489 the subspace stage is inactive so the difference isolates rank normalization. (**B**) Subspace-dimension sweep with the gate forced on: mean balanced accuracy across the 20 partitions (± SD) for single members $q \in \{4, 8, 16, 32, 64\}$ ($q \leq 32$ on UCI-489), the three-member ensemble (filled markers), and the rank-normalized full-space model of each cohort (dashed horizontal line).

Table 6. Partition sensitivity and ablations of VoxNeuro (Section 2.11). Each row pairs two models within the same outer subject partition (20 partitions, StratifiedKFold seeds 0–19; for models with the G-SMOTE branch the augmentation seed is derived from the partition seed, so partition and augmentation realization vary jointly) or, for the last analysis, within the same augmentation seed at the reference partition; values are means ± standard deviations across partitions or seeds, Δ is the paired difference (model A minus model B), and the last column counts strictly positive differences (ties noted where they occur). On UCI-489 the subspace stage is inactive, so VoxNeuro coincides with the rank-normalized full-space model and the normalization comparison is listed once; the forced-gate row applies the subspace stage there regardless. Differences smaller than 0.01 in magnitude are given with four decimals. The subspace branch generates no synthetic subjects, so model B of the last analysis is fixed. Comparisons with the remaining standard comparators are archived with the code.

Analysis (A vs. B)	Cohort	Metric	Model A	Model B	Δ (A − B)	Δ > 0
VoxNeuro vs. z-standardized full-space model	UCI-489	BA	0.853 ± 0.019	0.853 ± 0.013	+0.0006 ± 0.0213	9/20 (4 ties)

Analysis (A vs. B)	Cohort	Metric	Model A	Model B	Δ (A − B)	Δ > 0
	UCI-489	macro-F1	0.852 ± 0.019	0.851 ± 0.014	+0.0006 ± 0.0222	11/20
	PD-252	BA	0.784 ± 0.017	0.721 ± 0.020	+0.062 ± 0.018	20/20
	PD-252	macro-F1	0.785 ± 0.014	0.734 ± 0.019	+0.050 ± 0.017	20/20
VoxNeuro vs. rank-normalized full-space model	PD-252	BA	0.784 ± 0.017	0.750 ± 0.016	+0.034 ± 0.017	19/20
	PD-252	macro-F1	0.785 ± 0.014	0.751 ± 0.018	+0.033 ± 0.017	20/20
Full-space model: rank normalization vs. z-standardization	PD-252	BA	0.750 ± 0.016	0.721 ± 0.020	+0.028 ± 0.023	19/20
	PD-252	macro-F1	0.751 ± 0.018	0.734 ± 0.019	+0.017 ± 0.023	17/20
VoxNeuro vs. full-space Euclidean-kernel ablation ($w = 0$)	UCI-489	BA	0.853 ± 0.019	0.851 ± 0.020	+0.0025 ± 0.0258	9/20 (2 ties)
	UCI-489	macro-F1	0.852 ± 0.019	0.849 ± 0.021	+0.0026 ± 0.0262	10/20 (1 tie)
	PD-252	BA	0.784 ± 0.017	0.752 ± 0.018	+0.031 ± 0.015	20/20
	PD-252	macro-F1	0.785 ± 0.014	0.744 ± 0.021	+0.041 ± 0.019	20/20
VoxNeuro vs. class-weighted logistic regression	UCI-489	BA	0.853 ± 0.019	0.828 ± 0.027	+0.025 ± 0.030	16/20 (1 tie)
	UCI-489	macro-F1	0.852 ± 0.019	0.826 ± 0.027	+0.026 ± 0.030	17/20
	PD-252	BA	0.784 ± 0.017	0.752 ± 0.024	+0.032 ± 0.020	19/20
	PD-252	macro-F1	0.785 ± 0.014	0.757 ± 0.024	+0.028 ± 0.018	19/20
VoxNeuro vs. PLS-DA	UCI-489	BA	0.853 ± 0.019	0.831 ± 0.024	+0.023 ± 0.034	12/20 (3 ties)
	UCI-489	macro-F1	0.852 ± 0.019	0.829 ± 0.025	+0.023 ± 0.035	13/20 (1 tie)
	PD-252	BA	0.784 ± 0.017	0.742 ± 0.023	+0.041 ± 0.022	19/20
	PD-252	macro-F1	0.785 ± 0.014	0.741 ± 0.021	+0.043 ± 0.020	20/20
Subspace stage forced on vs. gate rule	UCI-489	BA	0.842 ± 0.022	0.853 ± 0.019	−0.011 ± 0.021	4/20 (4 ties)
	UCI-489	macro-F1	0.840 ± 0.022	0.852 ± 0.019	−0.011 ± 0.022	6/20 (1 tie)
Subspace branch with G-SMOTE vs. without (24 augmentation seeds)	PD-252	BA	0.797 ± 0.008	0.816 (fixed)	−0.019 ± 0.008	1/24
	PD-252	macro-F1	0.804 ± 0.008	0.811 (fixed)	−0.007 ± 0.008	7/24

4. Discussion

The central methodological contribution is the alignment of the evaluation unit with the clinical unit. Subject-disjoint validation asks whether a model generalizes to an unseen person. The Grassmann representation uses the dependence among repeated recordings rather than treating the recordings as independent replicates. Basis invariance removes arbitrary coordinate choices, while the Euclidean view restores the coordinate-level location and dispersion magnitudes that a pure subspace representation discards.

Three findings follow from the results. First, under a leakage-safe protocol with identical folds and normalization, VoxNeuro had higher point estimates than every standard comparator on both cohorts at the reference partition and, on PD-252, higher scores than each of them in at least 19 of 20 further partitions. On UCI-489 its margin over the full-space Euclidean-kernel ablation was small and not partition-consistent, so that cohort provides no robust evidence of improvement over the Euclidean view, and the Grassmann kernel term contributes modestly in a low-dimensional, balanced cohort. Second, the ablations associate most of the PD-252 margin with two fold-local changes: rank normalization, which was associated with higher distance variability and higher performance, consistent with a reduced influence of extreme coordinates, and the supervised-subspace branch, which increased chordal-distance variability where 753 descriptors and about 600 training recordings had driven the chordal distances toward their maximum (Section 2.7); the branch comparison bundles projection, ensembling, and the removal of oversampling. The complete configuration lifted specificity at the default threshold from 0.531 to 0.734. Within the subspace ensemble the Euclidean view carried most of the performance and the Grassmann term added small, partition-inconsistent increments, so the geometric representation should be regarded as a complementary rather than a dominant source of information. Third, the branch choice is associated with cohort dimensionality: on the balanced 45-dimensional UCI-489 cohort the same supervised stage lowered balanced accuracy, and the gate rule left the full-space model in place. The observed contrast is consistent with the proposed gate ($d > n_k^{\mathrm{rec}}$), but two cohorts do not establish the general validity of that threshold, which was formulated after both cohorts had been inspected. None of these counts is population-level inference about model superiority. With respect to the objectives of Section 1, the first was addressed analytically: Equations (19) and (20) formalize the difference between record-wise and unseen-person risks. The second was supported on PD-252 but not clearly on UCI-489, and the third associates the observed paired differences with normalization and the supervised-subspace branch. Whenever repeated recordings from previously observed people are easier to classify than recordings from unseen people, record-wise scores are optimistically biased in direct proportion to the probability that a validation recording belongs to an already-observed person. Subject-disjoint folds remove that term entirely. Record-wise and subject-disjoint evaluations target different prediction settings, so their reported scores are not directly comparable; Equations (19) and (20) describe the conditional-risk mechanism when recordings from previously observed people are easier to classify, and subject-disjoint scores answer the question that a screening deployment poses [20,22].

An evaluation-focused perspective also clarifies how these results should be interpreted in relation to the wider literature. Reported accuracies for voice-based PD detection vary widely across studies, and validation strategy is a recognized source of that variation [16,17]. Record-wise splitting remains common in secondary analyses of repeated-recording corpora, whereas replication-aware methods have also been developed for UCI-489 [29]. Balanced accuracy and macro-F1 were used because both cohorts were evaluated at the subject level and PD-252 had unequal class sizes. Both metrics weight the diagnostic classes symmetrically rather than rewarding majority-class predictions [66,67]. Fold-level standard errors are reported descriptively, and no significance test is claimed: five outer folds provide a descriptive summary of variability, not a basis for inferential comparison [25].

A separate research-stage web interface (Figure 9) demonstrates browser-based capture with a commodity microphone for three speech tasks (sustained vowel, read passage using an excerpt of the Rainbow Passage [69], and free speech), together with an illustrative longitudinal display. The application was not connected to either evaluated cohort and was not assessed for usability, calibration, or clinical performance. Figure 10 places the prototype in a proposed, rather than validated, deployment workflow and distinguishes the components evaluated here from capture and decision-support stages that were not; the evaluated models used only three repeated sustained-vowel recordings per subject, and constructing a common representation from heterogeneous application tasks remains future work. Such low-friction, repeated measurement is consistent with technology roadmaps for PD care [50,51]. The application is a research-stage prototype, not an autonomous diagnostic system or medical device, and no application-collected data were used in the analyses reported here. Prospective use requires explicit privacy, consent, security, accessibility conformance, calibration, and clinical governance procedures.

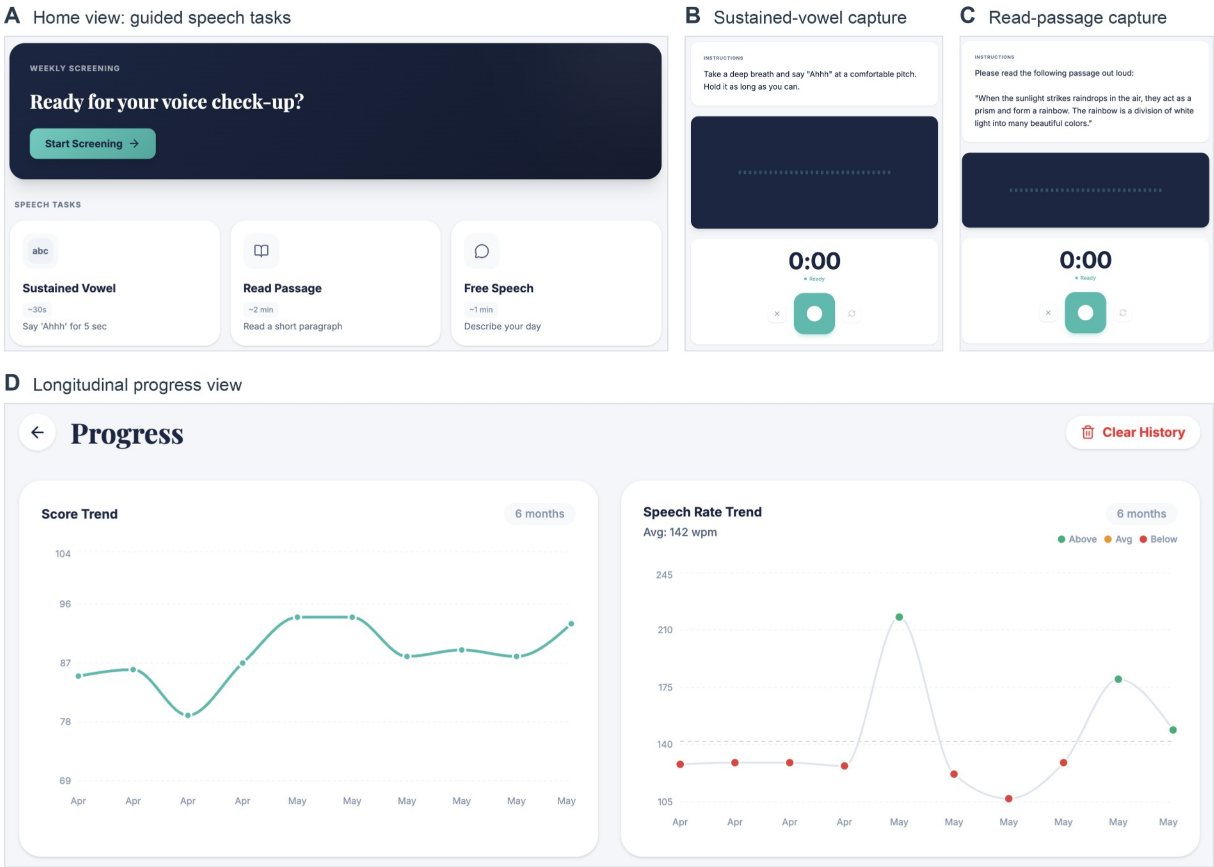

Figure 9. VoxNeuro web application (research-stage prototype, reachable at https://voxneuro.org after sign-in). (**A**) Home view listing the three guided speech tasks. (**B**) Sustained-vowel capture screen. (**C**) Read-passage capture screen showing an excerpt from the Rainbow Passage. (**D**) Longitudinal progress view with illustrative entries. The application requires only a commodity microphone and a web browser; no application-collected data were used in the analyses reported here.

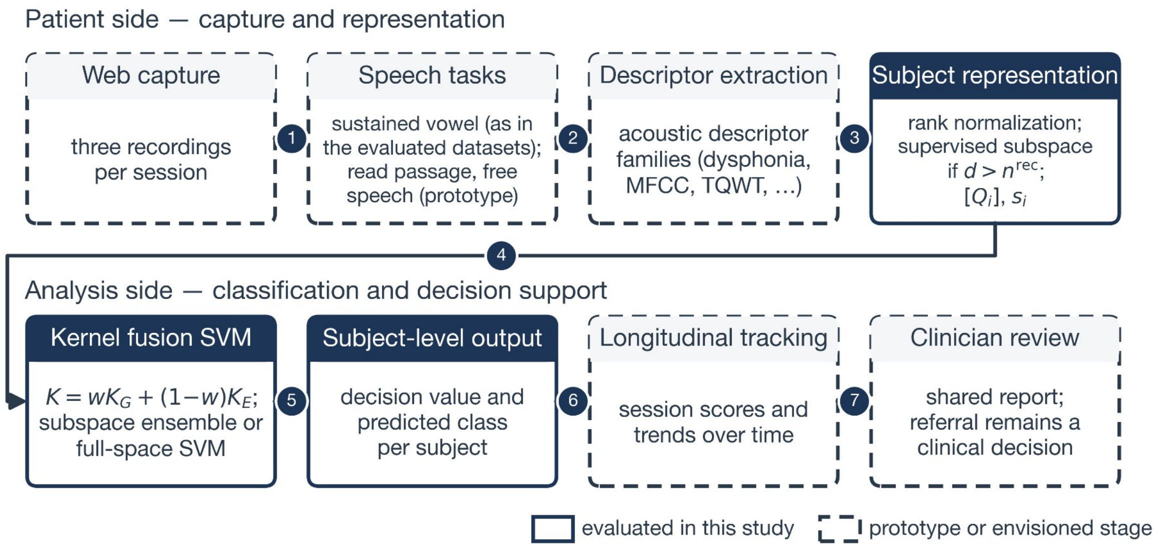

Figure 10. Study pipeline and envisioned deployment workflow of the VoxNeuro prototype. Solid boxes mark the components evaluated offline in this study on the two retrospective public cohorts (rank normalization and subject representation, with the supervised-subspace branch in the high-dimensional regime and the full-space branch with conditional G-SMOTE otherwise; kernel fusion; and subject-level scoring); dashed boxes mark capture and decision-support stages of the research-stage prototype that were not evaluated here, and no end-to-end pipeline from captured audio to classification was evaluated. Descriptor extraction in this study used the dataset-provided engineered features, and any referral decision remains a clinical judgment. The evaluated cohorts consist of three sustained-vowel recordings per subject obtained from the public datasets, not from the prototype's capture stage; the read-passage and free-speech tasks are prototype capture tasks that were not used in any evaluated model.

This study has limitations. The speech cohorts contain retrospective engineered features rather than prospectively acquired raw audio. The study included no probability calibration [70,71], decision-curve analysis [72], independent clinical test cohort, device-shift evaluation, language-shift audit, or prespecified subgroup analysis. Sensitivity to the fusion weight, the outer partition, the augmentation seed, the augmentation family, the subspace rank, the subspace dimension, and the normalization was characterized, but sensitivity to the SVM penalty, neighbor count, and median rule for the kernel coefficients was not, and the differences attributable to the geometric kernel term alone are small relative to fold-level variability. The final pipeline was chosen after exploratory comparison of several representations and fusion variants on the reference partitions of both cohorts, so its reference-partition estimates are subject to method-selection optimism (Section 2.11; the alternatives are documented in the project repository). The stability of the design choices was assessed only by repeated cross-validation on the same 80 and 252 subjects; the 20-partition results are not independent replications, and the gate rule has been exercised on exactly two cohorts. The PLS projection treats the three correlated recordings of a subject as separate observations when it is fitted, and the number of quantile knots, the subspace dimensions, and the equal-weight ensemble were fixed by convention rather than optimized. The comparator set did not include deep models, and pretrained audio models could not be evaluated because raw audio was unavailable. The Grassmann view alone was insufficient in both cohorts (balanced accuracy 0.600 and 0.636 under z-standardization at the reference partition; across partitions, 0.579 for the rank-normalized full-space model on UCI-489 and 0.636 for the subspace ensemble on PD-252), and the rank-two sensitivity analysis concerns the full-space branch only (Appendix A). The rank transform maps the binary gender covariate to two widely separated values, so its weight in the Euclidean view exceeds that of a continuous coordinate; its permutation effect was null under z-standardization (Appendix A.2), but its influence under rank normalization was not isolated. Because high PD-252 sensitivity coexists with modest specificity, a credible system should support referral and longitudinal review rather than replace neurological assessment. Future work should preregister preprocessing and thresholds and report in accordance with structured guidance for clinical prediction models [73,74]; collect repeated raw speech prospectively; and evaluate calibration, test–retest reliability, fairness, and clinician-facing decision consequences. The two-view construction extends naturally to additional recording tasks and views through further positive-semidefinite kernel terms, with weights selected by nested multiple-kernel optimization [42,43]. External validation of the frozen model on a cohort that played no role in its design is the decisive next step.

Reproducibility safeguards included fixed subject-disjoint folds, training-fold-only preprocessing, fixed hyperparameters, saved fold-level metrics, pooled confusion counts, and restriction of reported results to audited comparator implementations. The project repository archives, under a pinned dependency environment, subject-level out-of-fold predictions with fold identifiers for both cohorts, the comparator results under both normalizations, the partition analysis, the dimension sweep, the view and oversampling ablations, the concentration diagnostics, the supplementary analyses of Appendix A with the feature-family column map, and the scripts that generate them. The archived scripts and outputs permit regeneration and audit of the reported numerical summaries under the pinned software environment, and VoxNeuro is deterministic given the partition and single-threaded linear algebra.

5. Conclusions

VoxNeuro provides a leakage-safe framework for repeated Parkinsonian speech. Each subject is represented by a Grassmann subspace and a Euclidean mean–dispersion vector, the corresponding positive-semidefinite kernels are fused, and every learned component, including Gaussian rank normalization and, in the high-dimensional regime, a supervised-subspace ensemble, is fitted inside the training fold. Under one shared configuration the model reached balanced accuracies of 0.875 on UCI-489 and 0.816 on PD-252 at the reference partition and had higher point estimates than all five standard comparators (class-weighted logistic regression, a class-weighted linear support vector machine, SMOTE-based logistic regression, a class-weighted random forest, and PLS discriminant analysis) under identical folds and normalization. Across 20 further subject partitions it scored higher than every standard comparator on PD-252 in at least 19 partitions. Ablations associate the observed difference with rank normalization and the supervised-subspace branch, whereas UCI-489 provides no partition-consistent evidence of improvement over the Euclidean view. Clinically credible digital biomarkers require auditable implementations, transparent uncertainty, and evaluation designs that test generalization to unseen people; prospective external validation of the frozen model is the necessary next step before any clinical use.

Author Contributions: Conceptualization, L.K.; methodology, L.K.; software, L.K.; validation, L.K. and M.W.; formal analysis, L.K.; investigation, L.K. and M.W.; data curation, L.K.; writing—original draft preparation, L.K.; writing—review and editing, L.K. and M.W.; visualization, L.K.; project administration, L.K. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: Ethical review and approval were not required for this study, as it consisted exclusively of a secondary computational analysis of publicly available, fully de-identified datasets; no participants were recruited, and no new human data were collected.

Informed Consent Statement: Not applicable. The study analyzed publicly available, de-identified datasets and did not involve direct human participation; informed consent and ethics procedures for the original data collections are documented in the source publications cited in Section 2.1.

Data Availability Statement: The datasets analyzed in this study are openly available from the UCI Machine Learning Repository: the Parkinson Dataset with Replicated Acoustic Features (UCI-489; UCI dataset 489, doi:10.24432/C5701F) and the Parkinson's Disease Classification dataset (PD-252; UCI dataset 470, doi:10.24432/C5MS4X). Source code is available at https://github.com/khan-laiba/voxneuro (release v2.1.3); the repository also archives, under the pinned dependency environment, subject-level out-of-fold prediction files for both cohorts and both normalizations, the comparator results, the 20-partition analysis, the subspace-dimension sweep, the view and oversampling ablations, the concentration diagnostics, the supplementary analyses of Appendix A (augmentation seeds, augmentation family, outer partitions, rank, fusion-weight sweep, and permutation analysis with its feature-family column map), and the scripts that regenerate them. The research-stage web application is reachable at https://voxneuro.org (Google sign-in required; no application data were used in this study). No new primary data were collected in this study.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Supplementary Analyses of a z-Standardized Full-Space Variant

This appendix characterizes a z-standardized full-space variant of the model (fused kernel with $w = 0.5$ and the G-SMOTE gate) retained for sensitivity analysis. The results describe that variant and do not establish the behavior of the rank-normalized full-space branch or of the complete model; they are reported because they isolate properties of the fused kernel and of the geodesic oversampling branch that the complete model contains. All analyses used the protocol of Section 2.11 and are archived with the reference implementation.

Appendix A.1. Sensitivity to the Augmentation Seed

On PD-252, G-SMOTE randomly samples anchor subjects and neighbors and draws interpolation coefficients, so performance varies with the augmentation seed. To isolate this source of variability, the fixed five-fold evaluation was repeated for 24 distinct augmentation seeds while the outer-fold assignments, preprocessing, hyperparameters, and all non-augmentation settings were held fixed. Within each seed and fold, the fused model and the matched Euclidean-kernel ablation used the same augmented training set, so their difference is paired at the seed level. All PD-252 results of the full-space comparators use the reference implementation's default augmentation seed, which is archived and regenerable; this appendix characterizes variability around it for the z-standardized full-space model.

The paired analysis characterizes the sensitivity of the z-standardized full-space model, of its Euclidean-kernel ablation, and of their difference to the augmentation seed (Table A1); VoxNeuro generated no synthetic subjects on either evaluated cohort and is unaffected by this seed. Across 24 seed runs, balanced accuracy was 0.739 ± 0.012 for the fused model and 0.731 ± 0.013 for the matched Euclidean-kernel ablation, and macro-F1 was 0.752 ± 0.013 and 0.729 ± 0.013, respectively (mean ± standard deviation across seeds). The paired macro-F1 difference averaged +0.023 ± 0.015 and was positive in 23 of 24 runs. The paired balanced-accuracy difference averaged +0.009 ± 0.014, was positive in 16 of 24 runs, and reversed sign in 8 of 24. The archived default-seed realization of Table 4 (0.731 and 0.749) lies within the seed ranges [0.722, 0.760] and [0.732, 0.777]. Across the same 24 seeds, the fully Euclidean SMOTE plus Euclidean RBF pipeline averaged 0.726 ± 0.009 and 0.718 ± 0.010; replacing ordinary SMOTE by G-SMOTE in the Euclidean-kernel model changed balanced accuracy by +0.005 ± 0.016 and macro-F1 by +0.011 ± 0.016 (positive in 15 and 17 of 24 seeds), and the complete geometric pipeline exceeded the fully Euclidean pipeline by +0.014 ± 0.015 and +0.034 ± 0.016 (positive in 19 and 24 of 24 seeds). Thus, conditional on the fixed outer folds, the macro-F1 margin of the fused kernel over its Euclidean ablation was directionally consistent in all but one seed run, whereas the balanced-accuracy margin can be reversed by individual draws.

Table A1. Sensitivity of the full-space branch to the G-SMOTE augmentation seed on PD-252 (z-standardized full-space model versus its Euclidean-kernel ablation; Appendix A.1): paired evaluation across 24 distinct seeds with outer folds, preprocessing, and hyperparameters held fixed. Values are means ± standard deviations (SD) across seeds; Δ denotes the paired difference (fused minus ablation) computed within each seed, summarized by its mean ± SD, median [interquartile range, IQR], and sign count.

Metric	Fused	Ablation	Δ mean ± SD	Δ median [IQR]	Δ > 0
Balanced accuracy	0.739 ± 0.012	0.731 ± 0.013	+0.009 ± 0.014	+0.008 [−0.002, +0.018]	16/24
Macro-F1	0.752 ± 0.013	0.729 ± 0.013	+0.023 ± 0.015	+0.020 [+0.012, +0.034]	23/24

Each seed-level value is the five-fold cross-validation point estimate under fixed outer folds; standard deviations describe variation across the 24 augmentation seeds and are distinct from the fold-level standard errors of Table 4. Δ ranges: balanced accuracy [−0.013, +0.041]; macro-F1 [−0.001, +0.061]. Per-seed values are archived in the project repository.

Appendix A.2. Feature-Family Permutation Sensitivity

For feature family g, the fold-level decrease in performance and its summary are

$$\begin{aligned} I_{gkb} &= m_k - m_{kb}^{\pi_g}, \acute{I}_{gk} = \frac{1}{B}\sum_{b=1}^{B} I_{gkb}, \hat{I}_g = \frac{1}{F}\sum_{k=1}^{F} \acute{I}_{gk}, \\ \mathrm{SE}(\hat{I}_g) &= [\frac{1}{F(F-1)}\sum_{k=1}^{F}(\acute{I}_{gk} - \hat{I}_g)^2]^{1/2}. \end{aligned} \tag{A1}$$

Sensitivity of the Euclidean summary of the z-standardized full-space model was evaluated on each held-out fold without refitting. The coordinates of s_i associated with family g were jointly permuted across held-out subjects while Q_i and all other summary coordinates were held fixed. In Equation (A1), m_k is the original held-out macro-F1 and $m_{kb}^{\pi_g}$ is macro-F1 after the b-th permutation draw. $B = 20$ permutation draws were made per family and fold with a fixed random seed, the decrease was averaged over draws before the fold-level summary, and the six families follow the descriptor taxonomy of the source dataset (column assignments are archived with the code).

The TQWT summary block produced the largest mean decrease in held-out macro-F1, at 0.149 with an SE of 0.020, followed by the MFCC summary block, at 0.046 with an SE of 0.008 (Figure A1); within a fold, the standard deviation across the 20 permutation draws averaged 0.067 for TQWT. The vocal/time-frequency, wavelet, IMF/EMD, and demographic families produced mean decreases near zero. These exploratory values measure dependence of the fitted model on the Euclidean summary coordinates under correlated features; they do not establish biological causality and, because only the Euclidean view is permuted, do not quantify the dependence of the full two-view model on each family. The prominence of the TQWT family is consistent with the descriptor engineering of the source dataset, where tunable Q-factor features were introduced precisely for Parkinsonian voice [31,55].

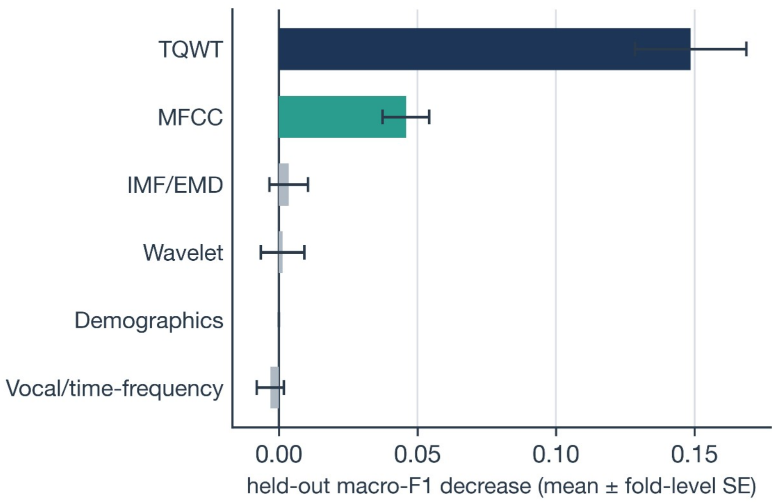

Figure A1. PD-252 feature-family permutation sensitivity of the Euclidean summary view of the z-standardized full-space model (Appendix A.2) under the archived default-seed realization (20 permutation draws per family and fold). Positive values denote a decrease in held-out macro-F1 after permutation, and whiskers denote fold-level standard error. TQWT denotes tunable Q-factor wavelet transform; MFCC denotes Mel-frequency cepstral coefficients; IMF/EMD denotes intrinsic mode function/empirical mode decomposition; Demographics denotes the dataset-provided gender covariate.

Appendix A.3. Partition, Rank, Augmentation-Family, and Fusion-Weight Sensitivity

Four supplementary analyses of the z-standardized full-space model were performed with the protocol of Section 2.11. First, the five-fold evaluation was repeated for 20 alternative stratified subject partitions (StratifiedKFold seeds 0–19) with a fixed augmentation seed, and paired differences between models were summarized across partitions. Second, the fused model was rerun with rank $r = 2$, for which every subject matrix has full rank, to test sensitivity to the rank choice and to the two rank-deficient subjects noted in Section 2.3. Third, on PD-252 the augmentation family was varied across the same 24 seeds as Appendix A.1, so that the fused model, the matched Euclidean-kernel ablation with G-SMOTE, and the fully Euclidean SMOTE plus Euclidean RBF pipeline were compared with paired draws. Fourth, the fixed fusion weight was varied over $w \in \{0, 0.1, \ldots, 1.0\}$ in every seed and partition run (Figure A2 displays $0 \leq w \leq 0.8$), and a nested variant selected $w \in \{0, 0.1, \ldots, 0.8\}$ inside each training fold by repeated (three times five-fold) inner cross-validation on balanced accuracy over the same grid, so that the outer held-out folds were never used for selection; the inner selection used representations normalized on the complete outer-training fold.

Across the 20 partitions, the z-standardized full-space model and its matched Euclidean-kernel ablation had similar mean estimates on both cohorts (paired balanced-accuracy differences +0.004 ± 0.013 on UCI-489 and −0.0005 ± 0.0163 on PD-252; macro-F1 +0.005 ± 0.013 and +0.008 ± 0.017), whereas the fused model retained a macro-F1 advantage of +0.028 ± 0.019 over the fully Euclidean SMOTE plus Euclidean RBF pipeline on PD-252 (18 of 20 partitions). Reducing the rank to $r = 2$ changed the full-space model by less than one fold-level standard error (UCI-489: 0.838 and 0.834; PD-252: 0.739 and 0.755); this analysis concerns the full-space branch only and was not repeated for the complete model. Figure A2 shows the sensitivity of the full-space model to the fusion weight: across the 24 PD-252 augmentation seeds, weights $w = 0.2$–0.7 exceeded the $w = 0$ ablation in macro-F1 in 22 or 23 runs, whereas $w = 0.1$ did so in 15 runs; the highest mean balanced accuracy occurred at $w = 0.3$ and the highest mean macro-F1 at $w = 0.7$, and on UCI-489 the highest observed means also occurred at larger weights ($w = 0.7$); nested selection of the weight inside each training fold produced positive balanced-accuracy differences against the Euclidean-kernel ablation in 20 of 24 seed runs and 13 of 20 partition runs, with mean differences of +0.0055 and +0.0007; these exploratory comparisons did not identify a preferred weight-selection procedure, and the archived variants of the full-space model (dimension-reduced, rank-one, centered, and block-balanced Grassmann views, decision-level fusion, weight ensembles, other penalties) are documented with the code without a claim of improvement.

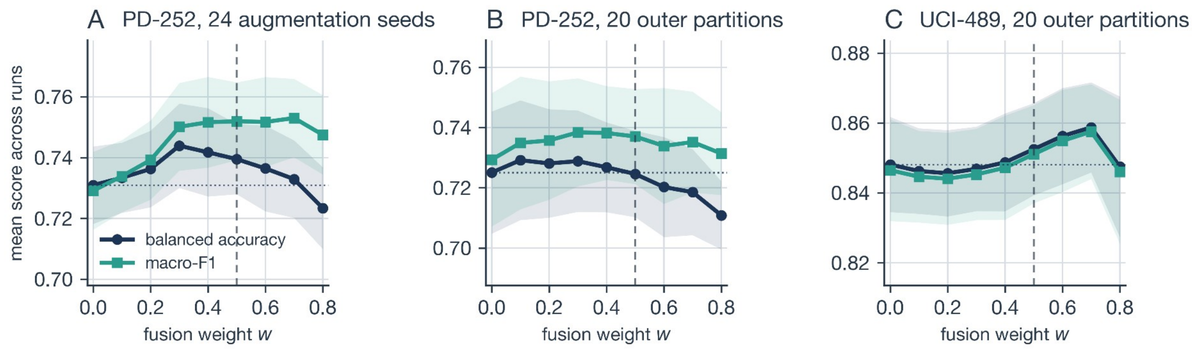

Figure A2. Sensitivity of the z-standardized full-space variant to the fusion weight w (Appendix A.3): mean balanced accuracy (navy circles) and macro-F1 (teal squares) across (**A**) the 24 augmentation seeds on PD-252, (**B**) the 20 outer partitions on PD-252, and (**C**) the 20 outer partitions on UCI-489. Shaded bands denote ± one standard deviation across runs, the dashed line marks the shared weight $w = 0.5$, and the dotted line marks the balanced accuracy of the matched Euclidean-kernel ablation ($w = 0$).

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